

WASP-135b

R-0175

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_250628 · created 2026-07-16T18:19:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460854.721 +/- 0.012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.041
Transit depth (Rp/Rs)^2	1.51 +/- 1.01 [%]
Orbital Inclination (inc)	86.92 +/- 2.66
Airmass coefficient 1 (a1)	1.035 +/- 0.021
Airmass coefficient 2 (a2)	-0.037 +/- 0.043
Scatter in the residuals of the lightcurve fit is	2.06 %
Best Comparison Star	#2 - [319, 277]
Optimal Aperture	2.66
Optimal Annulus	11.9
Transit Duration (day)	0.0845 +/- 0.0085

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.041 vs 0.139 (deviation 0.39σ)
Duration fit vs published	0.0845 d vs 0.0687 d (deviation 1.85σ)
Mid-time O–C	-3.5 min
Depth SNR	3.0
Residual scatter	2.06 %

Light curve

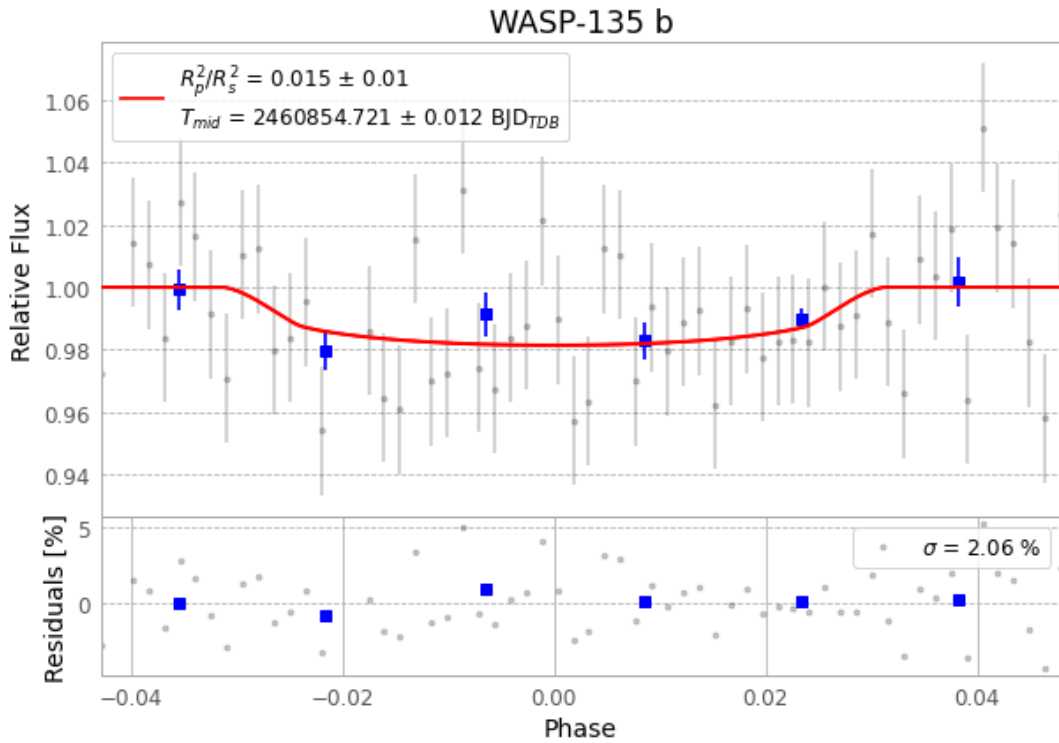


Figure 1 — FinalLightCurve_WASP-135 b_2025-06-27.png (SHA-256 04b8c5b4b8c75e52...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 198], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 72f7b32a09e7a465...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f67d25a

Data provenance

61 FITS frames (40 MB), WASP-135250628034513.FITS → WASP-135250628064816.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250628.log (f5ac9201e808...), FinalLightCurve_WASP-135 b_2025-06-27.png (04b8c5b4b8c7...), FinalLightCurve_WASP-135 b_2025-06-27.csv (86f29bc38680...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 18:19Z.